

Title

Perioperative esketamine reduces postoperative pain hypersensitivity compared with higher-dose opioid analgesia in emergency surgery

Abstract

Objectives: Post-surgical pain sensitization may drive chronic pain development. We evaluated whether perioperative sub-dissociative esketamine attenuates heightened pain sensitivity following emergency surgical intervention. **Methods:** Eighty-four patients undergoing urgent laparoscopic salpingectomy for ectopic pregnancy were randomized to intraoperative esketamine bolus (0.25 mg/kg) plus multimodal postoperative analgesia versus enhanced opioid analgesia. Pain sensitivity was quantified using a validated 17-item questionnaire administered on postoperative days 1, 2, and 7. Pain intensity, mood, and inflammatory markers were measured concurrently. **Results:** Pain sensitivity scores were significantly lower in the esketamine group compared with controls (main effect $F=9.88$, $p=0.002$). This benefit was consistent across both minor pain items ($p=0.002$) and moderate pain items ($p=0.002$). Acute pain intensity at 30 minutes post-recovery was lower with esketamine (median 3 versus 4 on numeric scale; 95% CI 0.57-2.42; $p=0.11$). Serum brain-derived neurotrophic factor concentration was elevated in esketamine recipients. **Conclusions:** Adjunctive perioperative esketamine modestly attenuates early postoperative pain hypersensitivity and may offer benefits within multimodal analgesic regimens for acute surgical pain.